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Student ID:

Course-Section: CSE425-11

Exam: Quiz-1

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Page No:

Serial No:

(Circle ... appropriate digits)

1	1	2	3	4	5	6	7	8	9	
0	0	1	2	3	4	5	6	7	8	9
	0	1	2	3	4	5	6	7	8	9
	0	1	2	3	4	5	6	7	8	9

1. What will be the output for the *query*: $m(X, [10, 20, 30]), write(X), nl, fail.$? Explain step-wise how Prolog arrives at the answer, including all unification attempts and backtracking. [marks: 6]

$m(X, [X | _]).$
 $m(X, [_ | T]) :-$
 $m(X, T).$

2. Trace the output of the query: $m(4,7,R)$ with and without the cut (!) predicate in the following code segment. [marks: 6]

$m(X, Y, X) :- X >= Y, !.$
 $m(_, Y, Y).$

3. Given the following Prolog program. What will be the output for the *query*: $enjoys(john, food).$? Explain step-by-step how Prolog arrives at the answer, including all unification attempts and backtracking. [marks: 8]

$likes(mary, food).$
 $likes(mary, icecream).$
 $likes(john, icecream).$
 $likes(john, mary).$
 $enjoys(X, Y) :- likes(X, Y).$
 $enjoys(X, Y) :- likes(X, Z), likes(Z, Y).$